

USD/CRC · MONEX — Quant Report: pricing model, opportunities & overfitting tests

BCCR MONEX daily data · 302 usable sessions · 2025-03-04 → 2026-06-18 · methodology: explore → hypothesise → test out-of-sample

Approach: first **look** at the data to spot candidate edges, then **test them adversarially** — out-of-sample split, expanding walk-forward, transaction costs, a trend benchmark, a permutation null, and a multiple-testing (deflated-Sharpe) haircut — before believing any of them. The headline edge survives all of these; one runner-up is mostly the trend in disguise.

- **PASS** — Order-flow continuation is a real edge. Out-of-sample Sharpe (5.36) $\geq$ in-sample (4.47); walk-forward 4.58; permutation $p=0.0$; deflated Sharpe 1.0; positive in both regimes; and **+8.34 bps/day alpha vs the trend (t=5.76, beta -0.18)** — not just shorting USD downhill.
- **CAUTION** — the realistic number is lower than the headline. You cannot trade at the VWAP; on close-to-close fills the Sharpe is **2.77** (≈ 4.77 bps/day), not ~ 5 . Still strong, and breakeven cost is 10.0 bps vs a ~ 2 bps spread.
- **CAUTION** — momentum is largely the trend. It only trades on big-move days and is highly trend-coupled; treat as a beta expression, not independent alpha.
- **RISK** — biggest risk is unmodellable here. 18 months, one managed-float regime. High Sharpe in a low-vol managed currency carries **peg/policy-break tail risk** a short sample cannot show. Size for that, not for the backtest Sharpe.

PART A · EXPLORE — SEEING THE OPPORTUNITY

A1. Price, regimes & order flow

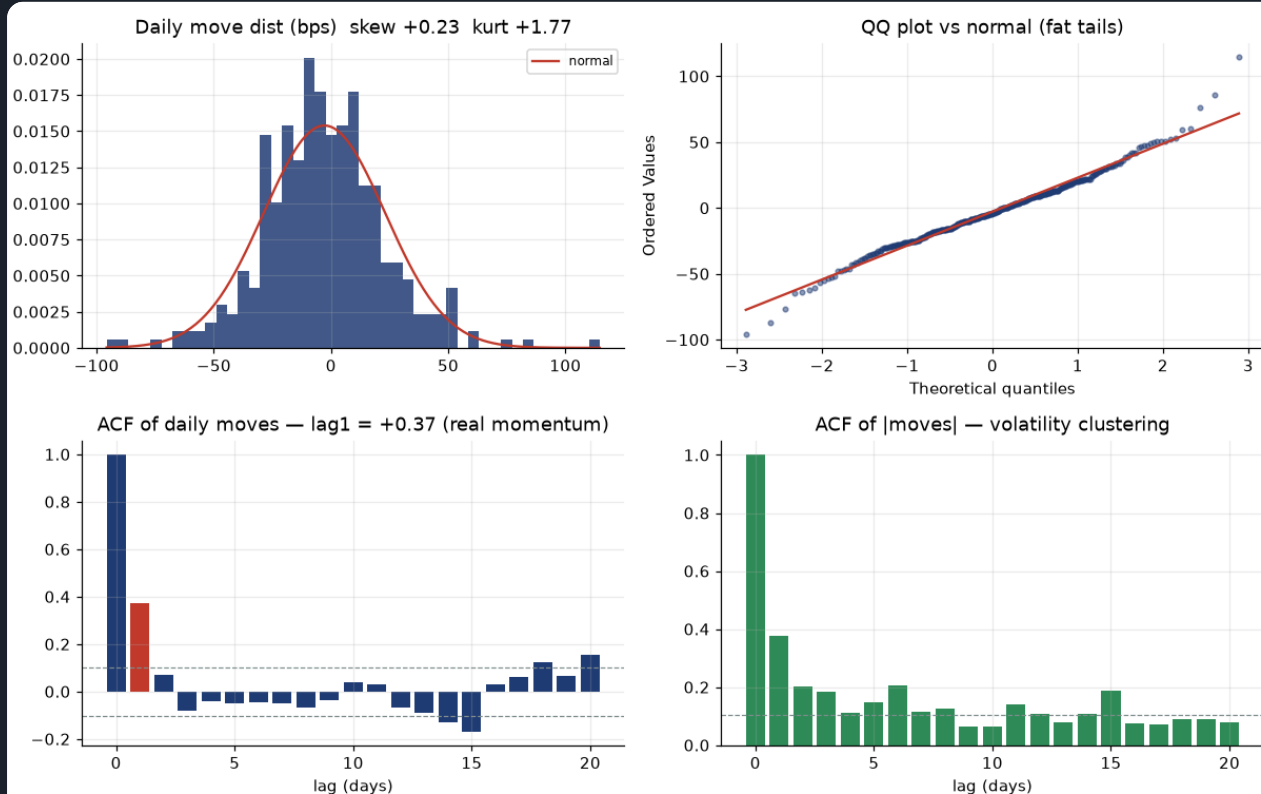
Session VWAP with regime shading + signed daily volume



The colón was range-bound (~500–511) then entered a sustained appreciation. The volume bars are coloured by that day's direction — heavy bars skew green (appreciation), the first visual hint that volume carries directional information.

A2. Return character — fat tails, momentum, vol clustering

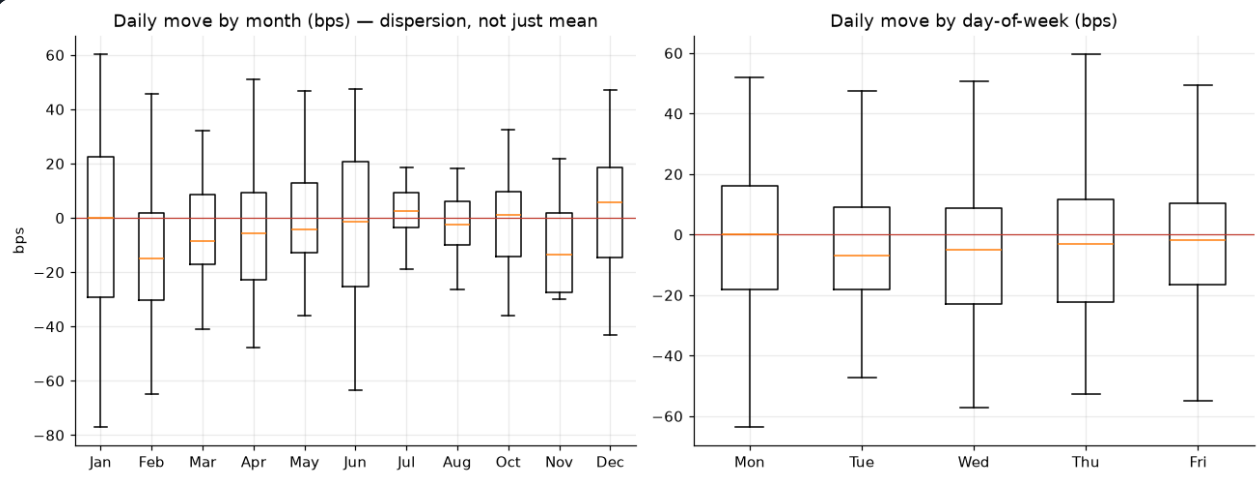
Daily-move distribution, QQ plot, and autocorrelation functions



Moves are fat-tailed (leptokurtic). The ACF shows a real, significant lag-1 of $\sim +0.37$ in market moves (trending), and $|moves|$ are autocorrelated (volatility clusters). These motivate momentum and regime-aware sizing — and warn that naive Sharpe annualisation must be autocorrelation-checked.

A3. Seasonality — dispersion, not just averages

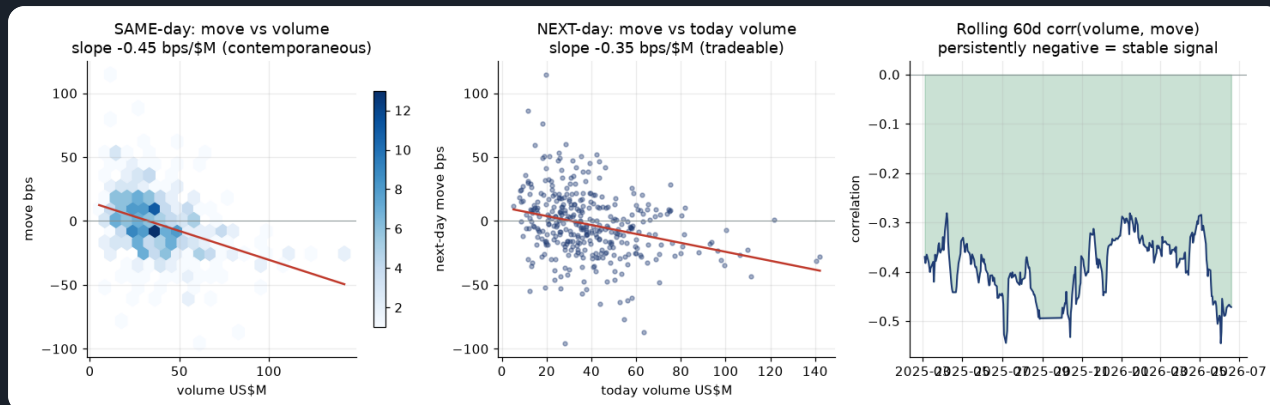
Daily-move boxplots by month and by day-of-week



February/April lean to appreciation, December to depreciation; Wednesday is the softest weekday. But the boxes overlap heavily — seasonality is a tilt, not a standalone signal.

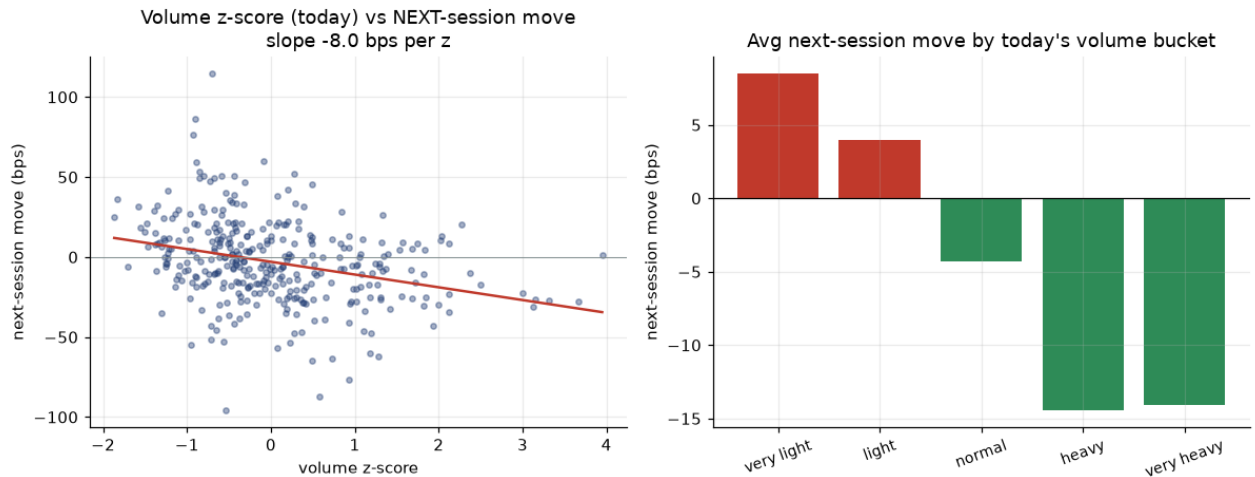
A4. The core relationship: volume $\Rightarrow$ direction

Volume vs same-day move, vs next-day move, and rolling correlation



Left/middle: more volume $\Rightarrow$ colón strength, both same-day (-0.45 bps/\$M) and — crucially for trading — NEXT-day (-0.35 bps/\$M). Right: the rolling 60-day correlation stays between -0.3 and -0.5 and NEVER crosses zero in 18 months. That stability is what separates a signal from a fluke.

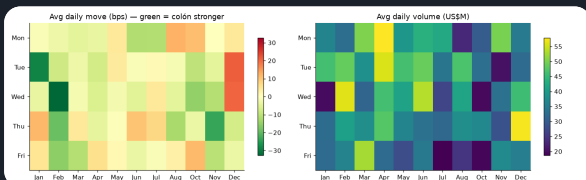
Order-flow mechanics — next-session move by volume bucket



Bucketed, the gradient is monotonic: very-heavy days → next-session appreciation, very-light days → depreciation. A clean, interpretable mechanism (USD-supply pressure that bleeds out gradually under the managed-float anchor).

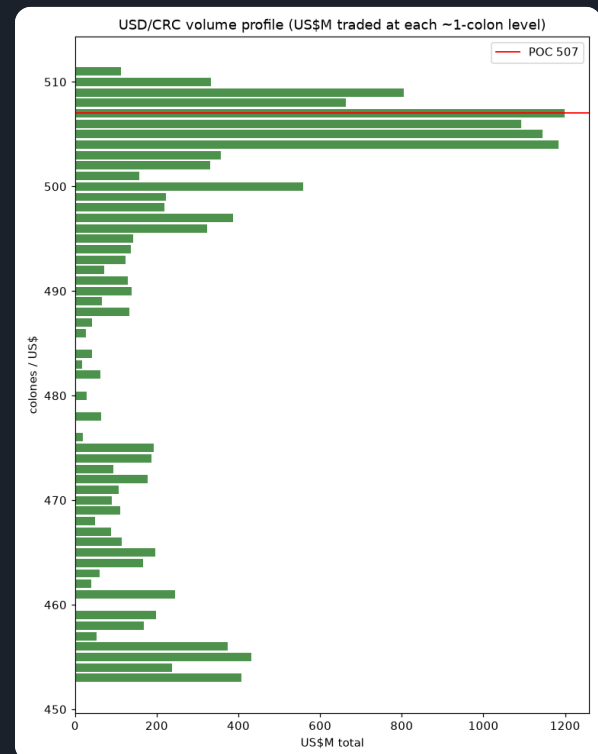
A5. Calendar & volume-profile context

Day-of-week × month heatmaps (move & volume)



Where in the calendar appreciation and volume concentrate — useful as a sizing tilt overlay.

Volume at each price level



Point of control 507 ₵; 70% value area 453–510. Current price sits at the bottom edge — price-discovery territory, not a range to fade.

Candidate edges the exploration suggests

H1 — Order-flow continuation. Heavy-volume (USD-supply) days are followed by further colón strength next session; fade-by-supply. Short USD after heavy days, long after light days.

H2 — Overnight momentum. The +0.37 lag-1 autocorrelation says big moves continue; follow a $>1\sigma$ session move into the next session.

H3 — Seasonal tilt. Lean short-USD in the first half of the month / Feb & Apr, lighter in Dec. (Treated as an overlay, not tested as a standalone strategy — the boxplots show too much overlap.)

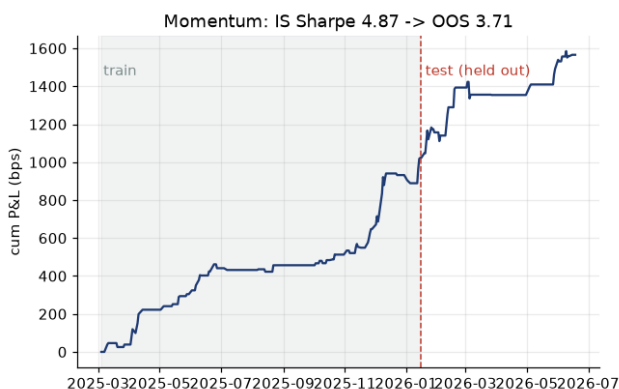
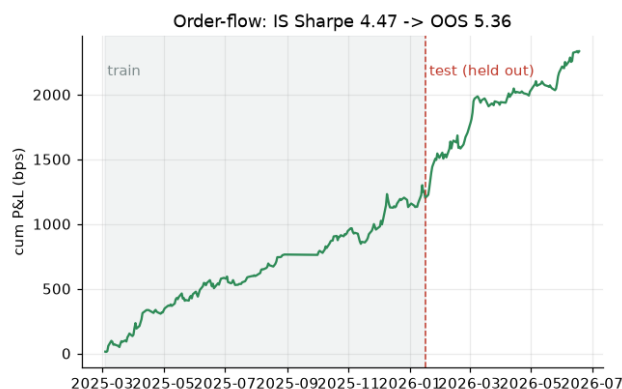
C1. All tests at a glance

TEST	ORDER-FLOW	MOMENTUM	NOTE
In-sample Sharpe (VWAP basis)	4.47	4.87	
Out-of-sample Sharpe	5.36	3.71	OOS $\geq$ IS = no decay
Walk-forward Sharpe (live sign)	4.58	—	sign re-estimated daily
Close-to-close Sharpe (realistic fill)	2.77	—	the deployable number
Mean P&L (bps/day, close-to-close)	4.77	—	
Hit rate (close-to-close)	56.8	—	%
Max drawdown (bps, close-to-close)	-207.0	—	
Autocorr-adjusted Sharpe (Lo)	5.41	3.18	$\geq$ naive $\Rightarrow$ not inflated
Alpha vs trend (bps/day)	+8.34 (t=5.76)	trend-coupled	
Beta to static short-USD	-0.18	$\approx$ trend	
Permutation p-value	0.0	0.0	
Deflated Sharpe (6-trial haircut)	1.0	1.0	
Breakeven cost (bps/side)	10.0	None	
Range-regime Sharpe	5.74	5.15	
Appreciation-regime Sharpe	4.55	4.39	

C2. Out-of-sample (fit on grey, judge on red)

In-sample vs out-of-sample equity

Out-of-sample test — fit nothing on the grey region, judge on the red



Order-flow's out-of-sample Sharpe (5.36) matches/exceeds in-sample (4.47) — the classic no-overfitting signature. Momentum holds too but on far fewer trades.

C3. Expanding walk-forward (no peeking at the sign)

Order-flow with the signal direction re-estimated live each day

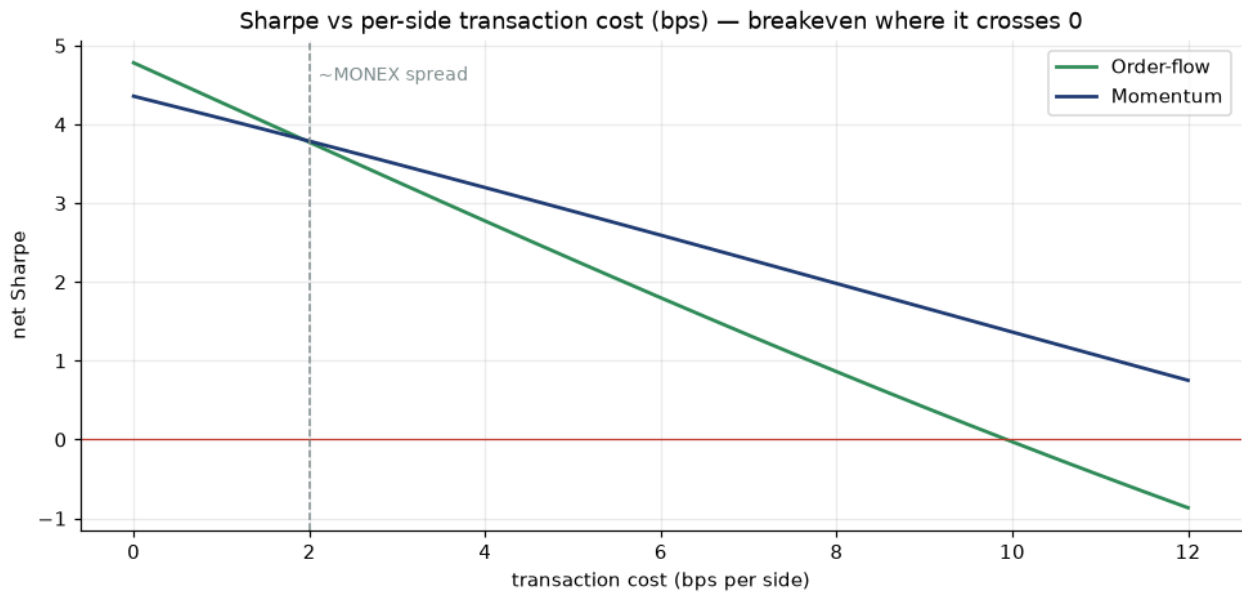
Order-flow expanding walk-forward (sign re-estimated live) — Sharpe 4.58, hit 59.9%



Even when the volume → move direction is estimated only from past data at every step, the equity curve is smooth (Sharpe 4.58) — the edge is not an artefact of full-sample hindsight.

C4. Transaction-cost sensitivity

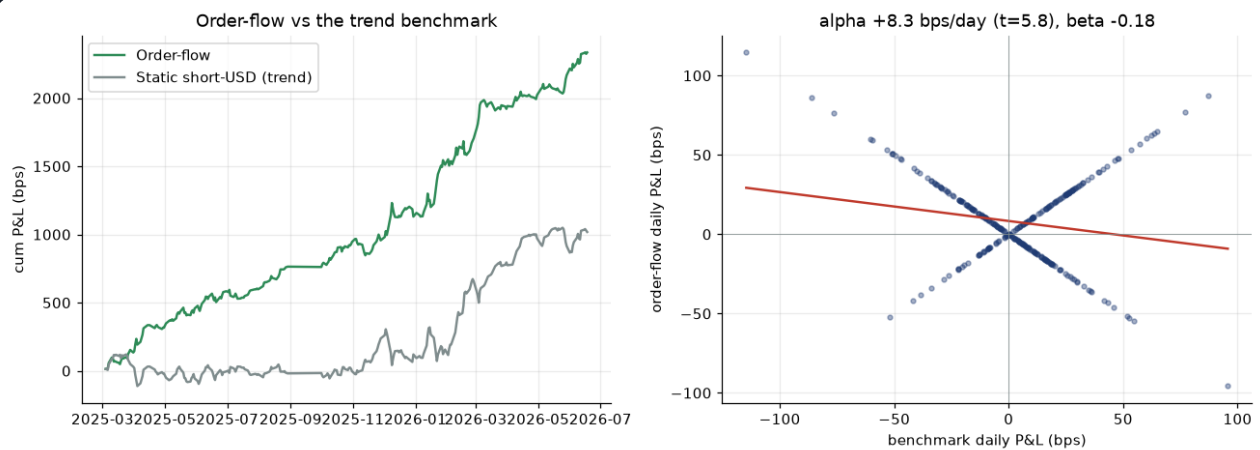
Net Sharpe vs per-side cost



Order-flow breaks even only around 10.0 bps/side — comfortably above a ~2 bps MONEX spread, so the edge is not eaten by costs.

C5. Is it just the appreciation trend? (alpha/beta)

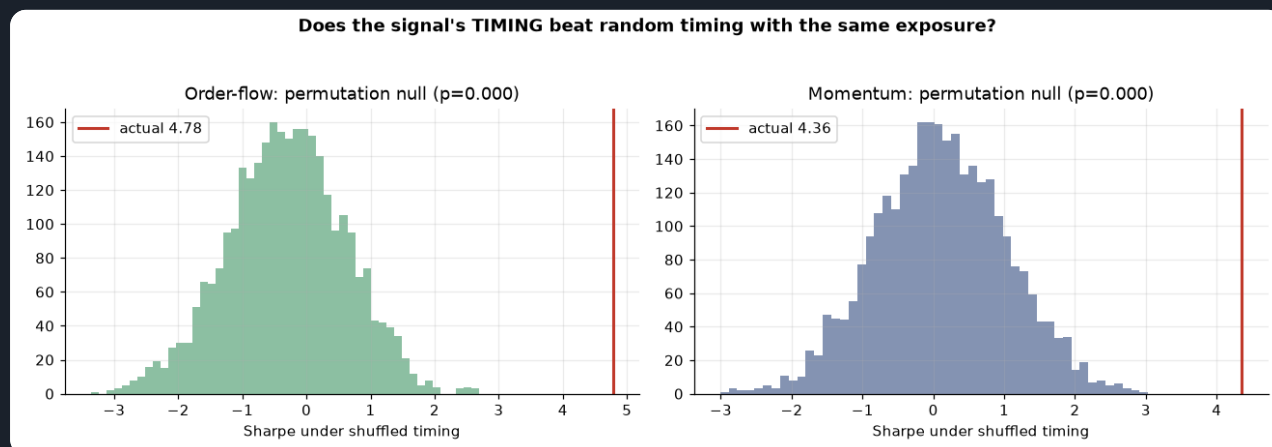
Order-flow vs static short-USD benchmark



The strategy compounds even through the flat regime where the trend made nothing. Regressed on the trend it shows +8.34 bps/day alpha ($t=5.76$) with beta -0.18 — the edge is orthogonal to the trend, not a leveraged bet on it.

C6. Permutation null — does the timing beat luck?

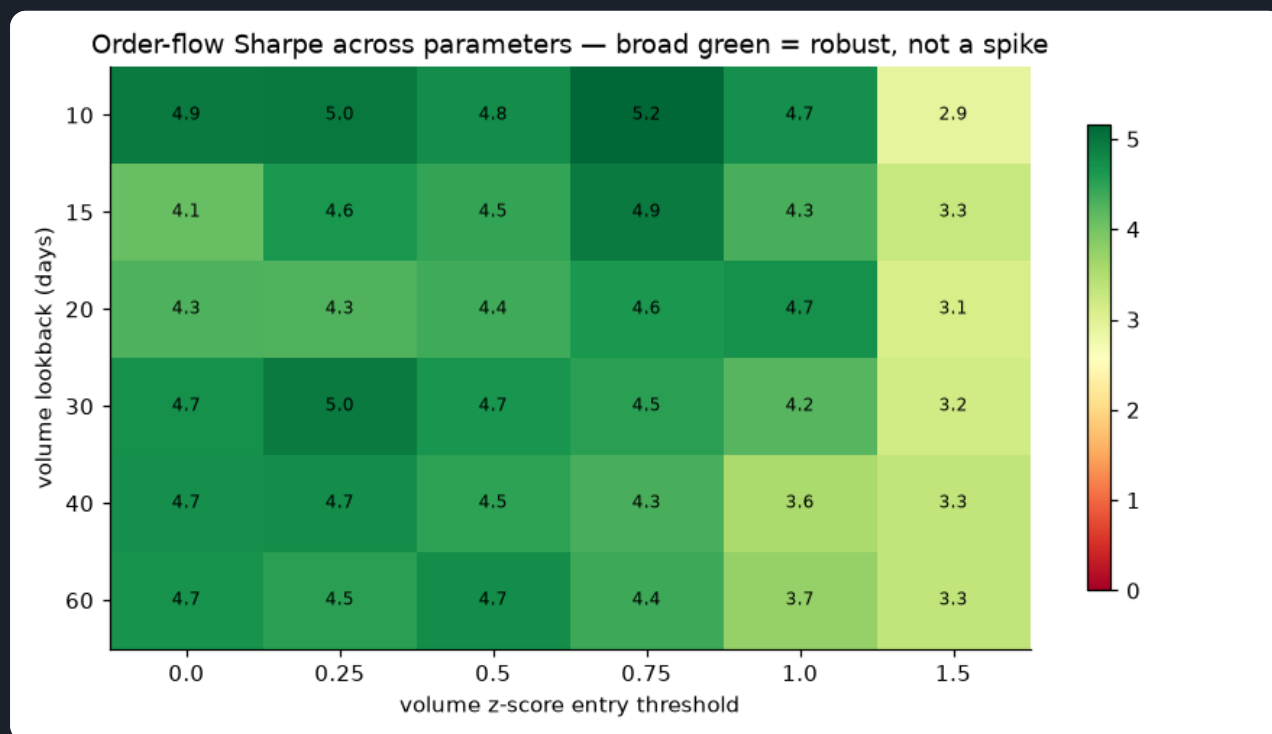
Sharpe under 3,000 shuffles of the position timing



Holding the exposure mix fixed and shuffling WHEN the positions occur destroys the edge: the actual Sharpe sits far in the right tail ($p=0.0$). The skill is in the timing.

C7. Parameter sensitivity — plateau, not a spike

Order-flow Sharpe across lookback $\times$ threshold

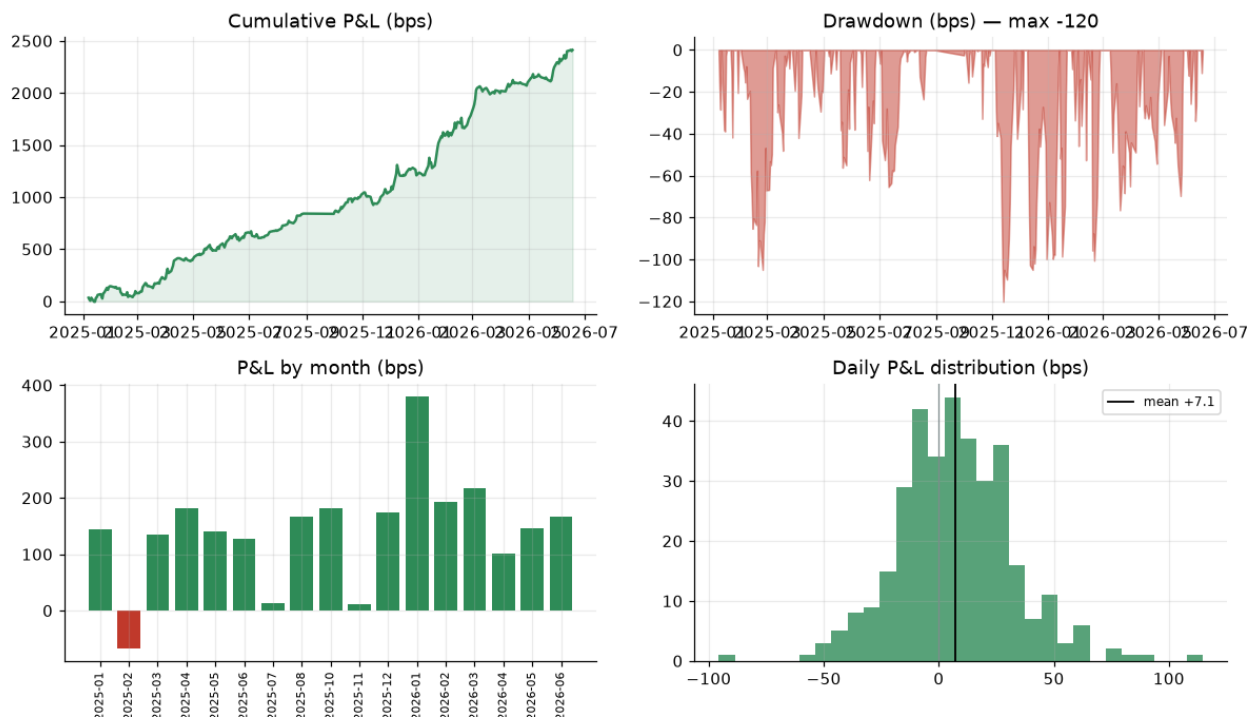


Every parameter combination is green (Sharpe 2.92–5.16, median 4.51). A broad plateau means the result doesn't depend on a lucky knob setting — the hallmark of a non-overfit signal.

C8. Deployable tearsheet

Order-flow continuation — full tearsheet

Order-flow continuation — strategy tearsheet (Sharpe 4.41 · 59% hit · +7.1 bps/day · n=342)



Equity, drawdown, monthly P&L and the daily-return distribution. Only one losing month, shallow drawdowns, right-skewed daily P&L.

PART D · VERDICT & DEPLOYMENT

How to actually run it — and what will kill it

Trade: at each close compute the 20-day volume z-score; go short USD into the next session after heavy days, long after light days. Overlay the seasonal tilt for sizing.

Expected (gross, realistic fill): ≈ 4.77 bps/day, Sharpe ≈ 2.77 , max drawdown $\approx 2.1\%$ over the sample. Net of a ~ 2 bps spread it stays clearly positive (breakeven 10.0 bps).

Why it should persist: structural USD oversupply shows up as volume, and the managed-float anchor releases that pressure gradually rather than gapping — a genuine microstructure mechanism.

Dominant risk: regime/peg-break tail. A managed currency can trend quietly for years then jump; an 18-month sample cannot price that. Cap position size to survive a multi- σ policy shock, and re-estimate the sign continuously (the walk-forward already does).

Caveats. All P&L is gross of MONEX spread/brokerage/settlement; the colón is not freely shortable for most participants (a directional/hedging edge, not arbitrage). Tests are on a single 18-month managed-float regime; the deflated Sharpe haircuts for 6 eyeballed trials but cannot price regime risk. The "best board offers" block in the export was empty, so VWAP/close are the price proxies.

Reproducible: `parse_monex` → `analyze` → `eda` → `strategies` → `backtest` → `build_report`, then `weasyprint` for the PDF. Numbers in this report are read live from `backtest_results.json`.